What a Recipient Can Recover from a Useful Prediction: An Evaluation of Feature Defenses and Output Contracts on Stored Representation Models

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Abstract

Methods that protect a declared attribute in a learned representation come with checks that cover a particular object and a particular class of attackers. A recipient served by a multi-purpose model may receive more than that object: a task output instead of, or together with, protected features, and two recipients that serve the same person may link what they receive. We ask what a recipient can recover about the attribute from everything it receives, alone or linked with a second recipient, and how useful the release remains. On stored Adult and HMDA representation models we apply each method's own check and then attack every released surface with one slate of attackers fitted on labelled releases of other records. LEACE meets its linear bound, yet nonlinear attackers recover the attribute about as well as from untreated features (AUC 0.817 against 0.815 on Adult and 0.863 against 0.870 on HMDA); the guarantee covers linear classifiers, so this recovery lies outside its scope. Because the untreated model's task output alone already reveals the attribute, releasing it beside noisy or FARE features brings recovery back to 0.76–0.79. When each treatment releases the output of a useful logistic-regression head, FARE lowers sex recovery from that score from 0.773 to 0.539, but a tree of the same budget without the fairness term accounts for about half of the reduction, and FARE's accuracy is not shown to stay within one point. In the one coalition we evaluated, our attackers recover more from two recipients granted perfect linkage than from either alone, even from decisions. All results, registered endpoints included, are development evidence on data used by earlier analyses, under a declared threat model; we propose no new method and no privacy guarantee.

1 Introduction

Methods that protect a declared attribute in a learned representation are stated and checked on that representation. LEACE erases the linearly available information about the attribute: on the distribution it is fitted to, no linear classifier reading the erased representation does better than a constant predictor under any convex loss [1]. FARE maps records to a finite set of cells and certifies, with high probability, a bound on the demographic-parity distance of any classifier that reads only the cell [12]. The purpose-specific encoders we study were trained with a penalty that targets a linear probe's R^2 for each disallowed attribute at or below a threshold. Each of these promises is precise about two things, the object it covers and the attackers it covers. Neither has to match what a recipient of the system receives or how the recipient can analyse it.

A release policy is defined by what each recipient receives. As an illustration, and not a description of any documented deployment, consider a service that holds records about people and serves two recipients with different permitted purposes: one receives a prediction of income, the other a prediction of an occupation group, and neither may use a declared attribute such as sex or race. For each person it serves, a recipient receives the output of its own task (a score, a probability vector or a decision) and possibly the features that produced it. The policy is written per recipient, but the person is shared. A recipient may know whom an output describes, so two recipients that serve the same person could link their outputs by record, and a recipient that knows the protected attribute for some of the people it serves could learn how outputs relate to that attribute. Under these conditions a defense applied to the features addresses only part of what recipients see: the task output may carry what was removed from the features, a recipient may hold features and output together, and two recipients may pool linked outputs. We know of no source that documents a deployment with all of these properties, so we do not claim that one exists.

These conditions raise a narrow question, which this paper studies. For stored representation models and published defenses, what can a recipient recover about a declared protected attribute from everything it receives, alone or linked with one other recipient, and how useful is that release for the recipient's task? We answer it under a declared threat model (Section 3). The recipient holds exactly its release, the attacker learns from a labelled sample of other records' releases from the same population, and a pair of recipients is granted perfect linkage. Other access, such as a recipient that also holds a person's inputs, public model parameters or permission to query repeatedly, defines a different threat model, which we do not test and do not dismiss.

We study six stored purpose-specific encoders trained on Adult and HMDA. Across them, we first apply each method's own protection check and then measure recovery from the surfaces a recipient receives under the evaluated contracts (Table 1): the representation; the task output in four formats (full logits, centred logits, probabilities and hard decisions); features and output together; and the outputs of two recipients linked by record. The same slate of linear, tree and neural attackers, fitted and selected on held-out roles, attacks every surface. Recovery is read beside held-out utility, which in Experiments 3 and 4 is mostly the utility of the head whose output is released and otherwise that of a refitted probe (Section 5). A tree of the same budget as FARE's but without its fairness term separates the effect of that term from the effect of compression.

The results separate three situations that a single recovery figure would conflate. A method can pass its own check while recovery continues outside the check's scope: LEACE meets its linear bound on every map, and nonlinear attackers recover the attribute about as well as from untreated features. The encoders' own constraint is a different case, because their stored linear check fails on 11 of 42 combinations of purpose–attribute pair and encoder seed. Recovery can also come through a surface that a feature defense does not cover: the original output alone recovers about 0.77, more than a label-only reference does, so a defense applied to the features cannot be credited with protection when that output accompanies them. Low recovery has to be read together with utility. The frozen task heads' hard decisions leak little, but they barely beat a constant. When every treatment releases a useful head, FARE's score recovers much less than the untreated or LEACE score, yet about half of that reduction is reproduced by a same-budget tree without the fairness term, and on an easier task the tree reproduces most of it. In the one coalition we evaluated, two recipients with perfectly linked outputs recover more than either alone, even when each receives only decisions.

None of this shows that LEACE or FARE fails its own promise: LEACE's held wherever we could check it, and FARE's certificate was too weak at our certification sizes to be checked. The encoders' stored check is the exception. What the results show is what a method must address if it aims to limit what recipients can learn: the complete release under a stated access model, the utility of the output that is actually released, a capacity-matched control for any effect claimed for a fairness term, and the linked view of recipients that serve the same people (Section 7). The individual observations, that linear protection does not stop nonlinear recovery, that outputs leak beyond labels, that hard labels are not a defense in general and that combined releases can leak more, are known (Section 4). This paper measures them together, with utility, on the same models and under one stated threat model, and adds two quantities specific to these models: the share of FARE's effect that a capacity-matched tree reproduces, which differs between settings, and, post hoc, the gap in recovery between useful scores and equally accurate decisions. Every assessment row had been used by earlier analyses of these models, so even the registered results are development evidence rather than fresh confirmation, and intervals are conditional on the fitted heads and attackers. We propose no new algorithm, no population privacy guarantee and no composition theorem.

2 Setting

Data. Adult is the UCI census-income data; HMDA is the 2023 California loan-application register (home purchase, first lien, site-built). Each dataset has three purposes, each with a task and a list of disallowed attributes. On Adult these are income prediction (two classes; disallowed race and sex), employment analysis (six occupation groups; race, age group, marital status) and education assessment (four education levels; sex, race, income). On HMDA they are underwriting (the loan decision; race, ethnicity), pricing analysis (five loan-amount bands; race, sex) and a fair-lending audit (whether the census tract's denial rate is high; race, sex). Together they give 14 purpose–attribute pairs. The primary settings are Adult income prediction

with protected attribute sex and HMDA underwriting with race; secondary analyses use all 14 pairs. Every experiment fits defenses and heads on the encoders' training split, without the 18 Adult and 43 HMDA training records that also occur in the test split (24,127 and 63,704 records), and divides the test split (15,060 and 13,661 records) into the attacker and assessment roles of Section 5. A class of the protected attribute is *supported*, and enters the macro AUC, if it has at least 100 records in the attacker's fitting role and in the assessment role and 30 in the attacker's selection role; the two smallest HMDA race groups are not supported.

Stored models. We use six stored purpose-specific encoders from earlier work (anonymized for review), one for each dataset and encoder seed (0, 1 and 2). Each encoder passes a record through a frozen, seeded, randomly initialised backbone, which is not pretrained, and a trained low-rank adapter for each purpose, producing a 64-dimensional representation per purpose; a frozen task head per purpose reads that representation. The encoders were trained with a linear constraint on each purpose's disallowed attributes: a one-hot ridge R^2 computed on each training minibatch, with the threshold 0.05 enforced through a Lagrangian penalty. Their stored check is a different object, an in-sample one-hot ridge R^2 on the test split with the same threshold. The HMDA encoder with seed 1 is collapsed: it maps many distinct records to identical representation vectors. The frozen heads of the two primary settings are weak predictors (Section 6.4), which is why Experiment 4 fits useful heads.

Heads and outputs. Three kinds of model produce task predictions. The *frozen head* is the task head stored with each encoder; it reads the untreated representation, and we call its full logits the *clean output*. An *own head* is a logistic-regression head that the owner fits on a treatment's features and whose only runtime input is those features (Experiments 2–4). The *probe* is a logistic-regression model refitted on the task labels of the attacker's fitting role and selected on its selection role; it measures utility in Experiments 1 and 2 and on Experiment 3's easier task, and its output is never released.

Recipients, release contracts and output formats. A *release contract* states what a recipient receives (Figure 1). Besides features alone, which we report for interpretation, a recipient receives (1) a task output only, from the frozen head in the frozen-head analyses of Experiments 1–3 and from an own head in Experiment 4, in Experiment 2's own-head surfaces and on Experiment 3's easier task; (2) features together with the output of their own head; or, as an adverse control, (3) features together with the clean output. The third path is a bypass that a release policy should not permit. An *output contract* fixes the format of a released output. For a K-class logit vector l, write the common offset $c = \frac{1}{K}\sum_k l_k$ and the centred logits $l - c\mathbf{1}$; for $K = 2$ the centred logits are determined by the margin $d = l_1 - l_0$. Probabilities and decisions do not depend on c (softmax shift invariance [7, §4.1]). The four formats are *full* logits, *centred* logits, *probabilities* and the *hard* decision. For the frozen heads, centred logits and probabilities carry the same information, full logits carry more and decisions less; for heads whose logits are log-probabilities, full logits, centred logits and probabilities are equivalent. Experiment 3 also releases the offset c alone and the pair (d, c), which determines the binary logits.

Feature treatments. A: the untreated representation. B: target LEACE [1], in its reference implementation (concept-erasure 0.2.4), which erases only the pair's protected attribute and is fitted on `defense_fit`, the owner's role on the encoders' training split. F: FARE [12], in the authors' implementation, whose release is the one-hot index of a tree cell. Its configuration comes from a frozen grid. Among the configurations whose probe accuracy on the attacker's selection role is within one point of the untreated probe's, the one with the lowest validation recovery for our nonlinear attacker is chosen. On HMDA the untreated probe's accuracy gain over the constant is only 0.6 to 2.3 points, so this rule admits a constant release: on HMDA seed 1 the chosen configuration has a single cell, keeps none of the gain and recovers 0.5 by construction, and on the other two HMDA seeds it has four cells. F0: the same FARE configuration with fairness weight zero, that is, a tree of the same budget grown on the task label without the fairness term, so that it only compresses. Experiment 1 also adds Gaussian noise to the representation, at the smallest scale σ^* whose validation recovery for our nonlinear attacker is at most 0.55 ($\sigma^* = 2$ on Adult and 4 on HMDA). FARE and noise were therefore tuned against the evaluated attacker slate on the attackers' own selection rows, which

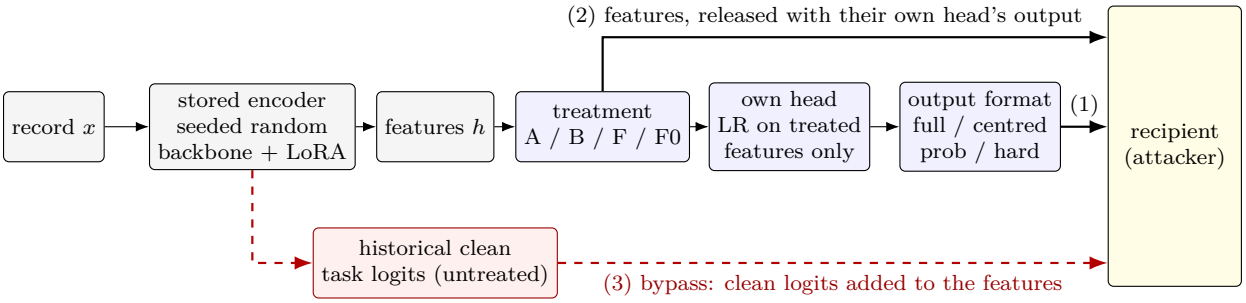


Figure 1: Recipient view. Solid paths are permitted releases: (1) output only, in one of four formats; (2) features of a treatment (A untreated, B LEACE, F FARE, F0 zero-fairness FARE tree) released with the output of their own head, which reads only those features. Path (1) is drawn for an own head; the frozen-head analyses of Experiments 1–3 release the frozen head's output, which reads the untreated features. The dashed path (3) adds the frozen head's clean logits to the features; it is the bypass, evaluated only as an adverse control.

can favour them when they are scored against the same slate; for noise, attackers outside the slate recovered more (Section 3). No defense grid was expanded and no configuration was re-selected after assessment.

3 Threat model and access

The recipient and its release. A *recipient* is a party permitted one purpose. For each record it serves, it receives that purpose's release under the contract in force and nothing else about the record: not the record's inputs, its true task label or its protected attribute, and not the model's parameters, gradients or internal states (defense maps, heads, or a representation that the contract does not release). Each record is released once, and no recipient submits inputs of its choosing. For the noise treatment, noise is drawn once per record for each of three release seeds; the three realizations are attacked separately and averaged, never pooled for one record. The label-only reference of Section 6.2 is a comparator, not a recipient: it reads the scored record's true task label.

Why this access. The claims we evaluate concern a release policy, so we chose to evaluate an adversary that holds what the policy delivers. That part of the access describes the service itself: an authorized recipient necessarily possesses the information delivered to it. Two further capabilities are granted by the evaluation and are not established for any deployment. The first is linkage. Two recipients that serve the same person may link their releases by record, and the linked-recipient evaluation asks whether protection that holds for each recipient survives when they pool. We grant perfect linkage, the attacker-favourable limit: a low pair recovery cannot be blamed on linkage error, while a high one may overstate what a coalition with imperfect linkage would recover. We do not assume that contracts or governance prevent linkage, and we do not claim that pooling is unavoidable. The second capability is labelled auxiliary data: releases of other records from the same population together with their protected attribute. This models a recipient that knows the attribute for some of the people it serves and can therefore learn how outputs relate to the attribute, as in attribute-inference work in which attackers train on labelled examples or on people who disclosed the attribute [11, 21]. The availability of such data, its size and its match to the scored population are assumptions of the benchmark; we do not show that a recipient could obtain such data in practice. Withholding the scored record's inputs isolates what the release itself reveals: with the inputs, recovery could come from imputation rather than from the release [10]. A recipient that holds inputs is a separate threat model, not an illegitimate one. More access cannot lower what an optimal attacker recovers, so a stronger attacker could recover more, but that alone does not justify granting more access. Each further capability, such as a defense's fitted parameters, defines a separate threat model that needs its own justification. A stronger attacker within the same access is different: a larger slate, or an attacker that uses public knowledge of a defense's algorithm and objective while seeing only the releases, belongs to this threat model. Our slate is neither, so each recovery figure is a lower estimate of what this access allows, not a guarantee.

Attackers. Our attackers are supervised. Each is fitted on the releases of the records in the attacker's fitting role, `attacker_fit`, labelled with the protected attribute; it is chosen by log loss on the labelled records of its selection role, `attacker_val`; and it is scored on `assessment` records that no attacker was fitted or selected on (Section 5). Tree and neural attackers also hold out an internal 10% of `attacker_fit` for early stopping. The sizes of the two labelled roles fix the attacker's data budget; they differ between experiments and are listed in Table 1. The labelled sample is a requirement of these supervised attackers, not of every conceivable attack, and attacks that need no labelled sample, for example ones built from a known population law, are not evaluated. Attackers are refitted to the releases of the defended system and so adapt to the distribution that the defense emits; only in that limited sense are they defense-aware. Knowing a defense's published algorithm and objective is different from holding its fitted parameters. A recipient may know the former. The slate's attackers, whose recovery the tables report, use neither beyond refitting to the defended releases; the one exception, run only in Experiment 1, is described below. Records are assigned to roles by a hash of a canonical record key, which also groups duplicate records; keys and row identifiers align releases and are never attacker inputs.

What recovery measures. Recovery is the held-out AUC that our attackers achieve. It describes a fixed attacker slate under this threat model, so a better attacker might recover more, and it is not a privacy guarantee. A difference in recovery between two treatments can reflect information about the attribute that one treatment removed, limits of our attackers at this data budget, or both; our measurements do not separate the two. A low recovery therefore does not show that the information is gone, and a high recovery does not show that all of it was kept. Being deterministic or real-valued does not make a release reversible: a function of the representation can merge records, and a deterministic map to probability vectors, which stays within a probability-output contract, can remove information about the attribute. We do not establish whether any evaluated release keeps the population distribution of the attribute given the record.

Access in each experiment. Table 1 gives, for each experiment, what the recipient receives, the labelled sample, how the attackers are fitted, selected and scored, what they know about the defense, how releases are linked, and what remains untested. The attackers of each experiment are described as they were implemented; no experiment is credited with a capability that only a later evaluation added. Experiment 1 also ran two exploratory attackers with more knowledge of the defense than the slate. On the noise treatment and the features-only surface only, it fitted two closed-form Gaussian likelihood-ratio attackers that use the noise law, an isotropic Gaussian with the selected scale σ^*, treated as a published parameter of the mechanism. The first is fitted, like the slate, on the labelled noisy releases of `attacker_fit`; it stays within the access above but uses knowledge of the defense's algorithm and scale. The second is fitted on clean, pre-noise representations of the `attacker_fit` records, which no contract releases, and so exceeds the access above. Neither is an insider or repeated-query attack, and neither has white-box access to the model: neither is selected on `attacker_val`, both score one noisy release per assessment record, and neither uses the encoder's or heads' parameters or gradients, or any assessment record's clean representation. At σ^* the first recovered 0.546 on Adult and 0.526 on HMDA, and the second 0.563 and 0.575, against 0.528 and 0.513 for the slate (exploratory values with unadjusted intervals and no paired test). The features-only noise entries of Table 2 are therefore not the strongest attack that was run on that surface; no noise-aware attacker was run on noisy features released with clean outputs.

Threat models outside this evaluation. The following are distinct threat models that we do not test. Several involve no breach of the provider: a recipient may legitimately hold a person's inputs, a model may be published, and a service may permit repeated queries. Some are governed by release-policy choices such as rate limits, one release per person, query auditing and whether to publish the model; others need security controls. We claim no protection against any of them and do not dismiss them: recipients that also hold a record's inputs or later learn its true label; public or leaked model parameters or gradients; permitted repeated, adaptive or chosen-input queries, including releases from several model versions; attacks that use a defense's fitted parameters, such as a LEACE map or a FARE tree; insiders; imperfect linkage; coalitions of more than two recipients; and auxiliary samples larger or smaller than ours, drawn from another population, or containing the auxiliary records' inputs or task labels. Within this access, attackers that

Table 1: Access in each experiment. Common to every attack of the slate: one release per record (noise: one draw per record for each of three release seeds, attacked separately); attackers fitted on labelled `attacker_fit` releases, selected by `attacker_val` log loss and scored on `assessment`; no attacker receives a record's inputs, model parameters, gradients or defense maps, and record keys are never attacker inputs. The exploratory likelihood-ratio attackers of Experiment 1 are the exception described in Section 3: they use the noise scale, are not selected, and one is fitted on clean auxiliary representations. Labelled sample: rows of `attacker_fit` / `attacker_val`. The label-only reference reads the scored record's true task label and is a comparator, not a recipient.

Experiment	What the recipient receives	Labelled sample (Adult; HMDA)	Defense knowledge, linkage, untested access
Encoders' stored check (a check, not an attack; reproduced in Exp. 1)	ridge fitted and scored in sample on the 64-dimensional representation of every test record	none; uses the scored records' own labels (15,060; 13,661)	the method's own test; covers only linear prediction from the representation
1. Feature removal (untreated and LEACE: all 14 pairs; noise: reported for the two primary settings)	representation (untreated, LEACE, or noise at σ^* on the primary settings); representation with the clean output; the clean output only	7,571 / 2,239; 6,794 / 2,089	slate refitted to releases; σ^* chosen on `attacker_val` with the slate; exploratory likelihood-ratio attackers know σ (see text). Features and output held by one recipient. Untested: repeated or fresh-noise queries, own-head outputs, other formats, pairs, and noise on eight of the twelve secondary pairs (a budget rule stopped the run; the four Adult pairs that ran are not reported)
2. Finite releases and output formats	FARE or zero-fairness cell (one-hot) or representation; either with the clean output; frozen-head output as full logits, probabilities or hard decision; features with the output of a head fitted on them, and that output alone (reused in Exp. 4); noisy features with each output format (not reported)	6,065 / 2,235; 5,388 / 2,082 (after removing 1,500 Adult and 1,385 HMDA rows for FARE certification)	slate refitted to releases; FARE configuration chosen on `attacker_val` with the slate; a cell-conditional attacker on some finite releases (descriptive). No pairs. Untested: attacks using FARE's tree or objective, noise-aware attacks, repeated queries
3. Output diagnosis	frozen-head output in six formats (full, centred, probabilities, offset, centred with offset, hard) and two format banks (Section 5), and the same formats for a refitted logistic head on the primary settings; Adult income and occupation-group outputs concatenated for one linked pair; features, own output or both on an easier task (Section 6.6)	as Exp. 2 (pair and easier task: Adult only)	no defense on the frozen and pair outputs; FARE configuration on the easier task chosen on `attacker_val` with the slate, among configurations that also keep 80% of the probe's gain; a cell-conditional attacker on hard outputs and FARE cells (descriptive). Pair: perfect linkage granted by aligned rows, linked labelled sample, single-recipient attackers among the candidates. Untested: imperfect linkage, more than two recipients, pairs that include features or defended outputs, HMDA pairs
4. Useful heads (Adult income / sex)	output of a logistic-regression head fitted on each treatment's features, in five formats (including the margin with the offset) and two banks; features with own output; features only; features with the clean output	as Exp. 2 (Adult)	slate refitted to releases; F is Exp. 2's configuration; a cell-conditional attacker on F and F0 outputs and hard decisions (descriptive). No pairs. Untested: repeated queries, other head families, HMDA

exploit a defense's published algorithm or objective were not run, apart from the exploratory noise-aware attackers of Experiment 1, one of which also used the auxiliary records' pre-release representations.

4 Related work

Concept erasure and its evaluation. Linear concept erasure aims to stop linear classifiers from recovering a concept: INLP [17] by iterative nullspace projection, R-LACE [18] through a linear minimax game, and LEACE [1], which guarantees it in closed form, for the moments of the data it is fitted on, with the least squared change among linear erasers. Such guarantees do not extend to nonlinear recipients: INLP and R-LACE report nonlinear recovery after linear erasure, kernelized erasure does not transfer across kernels [19], and LEACE restricts its claim to linear adversaries. Adversarially trained representations show the same pattern: post-hoc attackers recover what the in-training adversary could not [3] and still read the attribute from adversarially censored representations [21], and independent post-hoc adversaries are a standard check [16]. Our LEACE measurements are a further instance.

Leakage through task outputs. Model-inversion and attribute-inference attacks exploit confidence scores [4, 5]; label-only attacks can match confidence-based ones [15], and masking confidences does not stop label-only membership inference [2]. Wang et al. [25] compare leakage from predictions with that implied by the true labels at matched task performance; we report a label-only reference in the same spirit. A multiclass log-linear head on a linearly guarded representation can still reveal the erased concept, although a binary one cannot [20]. Black-box attribute inference on training records rarely exceeds what an adversary with the same distributional knowledge achieves by imputation [10], so we report constant and label-only references. Coarsening outputs, for example rounding confidences [4], is an old countermeasure; releasing probabilities rather than logits leaves calibration [8] and every proper task loss unchanged.

Fair representations and trade-offs. When the label depends on the protected attribute, fair representations trade accuracy for protection [26, 27]. Stadler et al. [22] prove that, for finite discrete data with full support and a strictly positive label posterior, a representation that carries more information about the task label than a level γ cannot keep, for *every* attribute, the inference gain of an adversary who sees it together with the label, over one who sees the label alone, within that level. The violating attribute is constructed, and the authors note that leakage of a single attribute can sometimes be restricted. We report an analogue of their label-only adversary as a descriptive comparison, not a floor: in our setting the releases are computed from the inputs alone, so they can reveal less, or more, than the true label does. In Experiment 2 the frozen heads' full outputs revealed more than the label-only reference and their hard decisions revealed less (exploratory). LAFTR [14] bounds the unfairness of any downstream classifier only for an optimal adversary; FCRL [9] evaluates with a common downstream protocol. FARE [12] restricts the encoder to a finite partition and certifies, with probability at least $1 - \epsilon$, a bound on the demographic-parity distance of any classifier that reads only the cell; its authors note that restricted encoders lose information and compare fairness-unaware k-means encoders. We add a same-budget zero-fairness tree and measure attribute recovery from the complete release.

Adversary models and combined releases. Our recipient is not a new adversary. It is the auxiliary-data attacker of attribute inference, which learns from labelled examples to read a sensitive attribute from what a model exposes [21] or from a person's public data [11], applied here to released task outputs as well as representations. Combining independent releases is a classical route to breaches [6]; recipients of differently perturbed copies of the same data can pool them, and mechanisms exist under which the pool reconstructs the data, by linear least-squares estimation, no more accurately than the best single copy [13]; an attack that combines the predictions of a biased and a fair version of one model raises attribute inference on image benchmarks [24]; and collusion-aware mechanisms exist for sequential finite-alphabet releases [23]. We measure the case in which each recipient is permitted a different purpose, as a property of our attackers.

Position. In our reading of the closest work on concept erasure, fair representations, output leakage and combined releases, we did not find one protocol that applies, on the same stored models, each method's

native check, nonlinear recovery from representation and complete output, the released output's held-out utility, a capacity-matched zero-fairness control, and recipient pairs. FARE is closest on matched utility and compression, Wang et al. [25] on utility-qualified output leakage, and Tian et al. [24] on combined releases. This reflects a bounded search and is not a claim of priority.

5 Evaluation protocol

Experiments. The evidence comes from four experiments on the same stored models. They share the attacker slate and the endpoint discipline described below, but they were run separately, with their own registered endpoints, and they differ in roles (Table 1 gives each experiment's labelled budget) and in how intervals are computed. Experiment 1 compares untreated and LEACE representations on all 14 pairs, and reports noisy representations at σ^* for the two primary settings. Experiment 2 adds FARE and the zero-fairness tree, and compares the frozen heads' full logits with their hard decisions. Experiment 3 measures the effect of withholding the logit offset, the usefulness of the frozen heads, a two-recipient pair, and FARE's compression share on an easier task. Experiment 4 replaces the frozen heads with a useful head fitted on each treatment's features. Table 1 gives each experiment's access.

Roles. Experiments 2–4 use the same record-group roles: `defense_fit` (owner side: the encoders' training split without its 18 Adult and 43 HMDA records that also occur in the test split, 24,127 and 63,704 records, on which LEACE maps, FARE trees and task heads are fitted, with a 20% head holdout for head selection), `attacker_fit` and `attacker_val` (attacker fitting and selection), and `assessment` (scoring only: 5,243 Adult and 4,764 HMDA records). Records that also occur in the encoder's training split are excluded from every scored role (17 Adult and 42 HMDA test records), and a further 20% of `attacker_fit` groups is reserved for FARE certification. The owner also uses the attacker roles: the task labels of `attacker_fit` fit the common utility probe and those of `attacker_val` select it and decide which FARE configurations are admissible, while the protected labels of `attacker_val` choose FARE's configuration and, in Experiment 1's roles, the noise scale (Section 2). No sensitive assessment label selects a release, grouping, penalty, attacker or utility probe. Experiment 1 used the original roles (assessment 5,250 Adult and 4,778 HMDA records), which retained the overlapping records; its verdicts were stable when they were removed.

Attackers and recovery. Every reported surface is attacked by the same slate, with the access of Section 3: logistic regression (5 regularisation values), gradient-boosted trees (20 configurations) and an MLP (18), each fitted on `attacker_fit` releases labelled with the protected attribute. The nonlinear attacker is the better of the tree and MLP families on `attacker_val` log loss, refitted at three attacker seeds. Recovery is the nonlinear attacker's macro one-vs-rest AUC on assessment over supported classes (Section 2), averaged over attacker seeds, release realizations (noise only) and then encoder seeds. For releases that combine features and output, selection on validation also considers attackers that use only the features or only the output. For the recipient pair of Experiment 3, the candidates also include each single recipient's attacker, chosen on validation like any other candidate. For some finite releases (Table 1) we additionally fit a cell-conditional attacker, which estimates the attribute's distribution within each released cell; we report it descriptively; in Experiment 2, where both were fitted, it was within 0.003 of the nonlinear attacker. In Experiments 3 and 4 a *bank* is a set of release formats of the same output; for each encoder seed the format whose nonlinear attacker has the lowest validation log loss is scored. The full bank includes the full logits and the offset-free bank does not. The label-only and constant references, the cell-conditional attacker, the exploratory likelihood-ratio attackers of Section 3 and the encoders' in-sample check are not part of the slate.

Utility. In Experiments 3 and 4 we report the accuracy, balanced accuracy, minority recall, log loss and gain over the constant (majority class on `attacker_fit`) of the *deployed* head whose output is released (the frozen heads in Experiment 3, the own heads in Experiment 4). The probe measures utility in Experiments 1 and 2 and on Experiment 3's easier task, and Experiment 2 also reports the accuracy and log loss of the own heads.

Native checks. The encoders' stored check is a historical value, the in-sample one-hot ridge R^2 on the test split with threshold 0.05, which Experiment 1 reproduced on all 42 combinations of pair and encoder seed. LEACE's implementation bound is a whitened residual cross-covariance below its tolerance on `defense_fit`. FARE's is its demographic-parity certificate ($\epsilon = 0.05$) on the certification role (1,500 Adult and 1,385 HMDA rows).

Inference. Each endpoint is a mean over encoder seeds of a per-seed statistic. In Experiments 3 and 4, standard errors come from 1,999 paired multinomial bootstrap replicates over assessment record groups, shared by every attacker fit, and intervals are simultaneous two-sided 95% Bonferroni normal intervals over the endpoint family. All intervals describe the mean over the three stored encoders, conditional on the fitted heads and attackers; they do not cover the variation from retraining encoders, which is large (Figure 2). Experiments 1 and 2 used one-sided Bonferroni percentile bounds from 20,000 group-bootstrap replicates over endpoint families of 24 and 12 ($\alpha = 0.05/24$ and $0.05/12$), and the bounds in Table 2 from those experiments are of that kind. In every experiment an endpoint passes only if its lower bound exceeds its registered target; Experiment 1's level endpoints are read three ways, as established above or below their bar or unresolved. Utility targets are a non-inferiority margin of -0.01, a gain of 0.03, and the retention contrast $\mathrm{Acc}(\text{treatment}) - 0.8\,\mathrm{Acc}(A) - 0.2\,\mathrm{Acc}(\text{const}) > 0$, which requires a treatment to keep at least 80% of the untreated head's accuracy gain over the constant; the recovery contrasts of Experiments 2–4 use a target of 0.02 AUC. Experiment 1's registered endpoints compare levels with fixed bars instead (recovery with 0.55 AUC, the held-out linear statistic with 0.05 and probe accuracy with a one-point margin), and Experiment 3's frozen-head usefulness endpoints used a gain target of 0.01. An interval that excludes zero but not the target is reported as a smaller difference below the target, never as "no difference"; an endpoint is *not established* when its lower bound does not exceed the target, which is not evidence of no effect. Each experiment was *registered*: its protocol, code, input hashes, list of fits and endpoint families were committed to the anonymous repository, with timestamps, before its first fit, and an independent replay that does not import the analysis code recomputed every primary result. The encoders' stored check predates these experiments; Experiment 1 reproduced its values. Changes made after an experiment's runs had begun are recorded with its protocol. Two lock amendments preceded the fits they affected: a map-identifier fix in Experiment 2, made after its Adult run and before its HMDA fits, and the addition of the easier-task code in Experiment 3, made before those fits. Two changes followed results: a guard in Experiment 2's FARE certification code that identifies duplicate rows by record rather than by feature values, without which every certificate was unavailable; and a scoring repair to the easier-task comparison of Experiment 3, found by the independent replay, which applied the registered rule for candidates that ignore one part and moved that endpoint's lower bound from 0.0001 to 0.0019 without changing its verdict. Experiment 2's report script was also fixed after a crash, and its reporting records were repaired at the start of Experiment 3; neither changed an estimand, selection, role or verdict.

6 Results

Table 2 collects the main contrasts from Experiments 1–3. Section 6.5 reports the useful-head comparison, and Sections 6.6 and 6.7 the compression share of FARE's effect and the linked recipients.

6.1 What each method's own check establishes

Three outcomes have to be kept apart: a method failing its own test, its test holding while recovery occurs outside the test's scope, and recovery through a surface that the test does not cover. The encoders' constraint is an instance of the first. Their stored linear check fails on 11 of the 42 untreated combinations (14 purpose–attribute pairs, three encoder seeds each), including HMDA underwriting/race on two of three seeds, so for those combinations the encoders do not deliver the protection that their own test describes. LEACE is an instance of the second. It meets its bound on all 60 fitted maps (including 18 fitted for a set of attributes), and its held-out squared top canonical correlation ρ_1^2 is small (point estimates 0.0006 on Adult and 0.0001 on HMDA, with unadjusted 90% upper ends of 0.013 and 0.033, on one already-used held-out split). This is a measurement on one split, not a guarantee about future records, and on seeds whose `defense_fit`

Table 2: Main contrasts from Experiments 1–3 (means over three encoder seeds; the noise row also averages three release seeds). Parenthesised values are lower confidence bounds of registered contrasts: a one-sided Bonferroni percentile bound in Experiment 2 and lower ends of simultaneous two-sided Bonferroni intervals in Experiment 3 (Section 5). Other rows are registered levels (Experiment 1), levels underlying registered contrasts (Experiment 2), an exploratory level (the zero-fairness tree), descriptive per-seed points and a count.

Quantity	Exp.	Adult income / sex	HMDA underwriting / race
Encoders' own linear check fails (pair×seed, all 14 pairs)	1	11 of 42	
Nonlinear recovery, untreated features	1	0.815	0.870
Nonlinear recovery, LEACE features	1	0.817	0.863
Noise at σ^*, features only / plus clean output	1	0.528 / 0.778	0.513 / 0.758
Full logits vs hard decision (frozen head)	2	0.776 vs 0.513: +0.263 (0.249)	0.768 vs 0.505: +0.262 (0.254)
FARE vs LEACE features	2	0.550 vs 0.812	0.501 vs 0.863
Zero-fairness tree (same budget), features	2	0.645	0.608
FARE features plus clean output	2	0.788	0.768
Offset withheld: full bank minus offset-free bank	3	+0.072 (0.062)	+0.067 (0.060)
per encoder seed	3	0.079 / 0.064 / 0.072	0.058 / −0.0003 / 0.143
Frozen head: accuracy gain over constant (target 0.01)	3	0.039 (0.031)	0.016 (0.0098), not established

covariance is rank-deficient LEACE leaves directions outside its fitting support unchanged. Nonlinear recovery nevertheless changes by less than 0.01 AUC (Adult +0.002, HMDA −0.007, from the levels in Table 2; for HMDA the unadjusted 90% interval of the change excludes zero). The guarantee concerns linear classifiers, and this measurement shows that our nonlinear attackers still recover the attribute about as well as from untreated features; it is not evidence against the theorem. FARE's certificate could not be assessed in the same way. At the available certification sizes it was unavailable (the procedure could not run, for example because a cell was absent from one certification split), vacuous (a bound above 1) or weak (Adult bounds about 0.50), and its only informative bound (0.175, a secondary certificate over three race groups) belongs to a degenerate HMDA configuration with a single cell. These certificates exist only after the post-run repair of the certification code described in Section 5.

6.2 The task output as a separate surface

The third outcome concerns what a feature defense does not cover. Output-only recovery exceeds the label-only reference by 0.177 on Adult and 0.236 on HMDA in Experiment 1 (exploratory; unadjusted 90% intervals [0.166, 0.188] and [0.227, 0.245]), so the outputs carry information about the attribute beyond what the true label carries. When the clean output is released beside noisy or FARE features, recovery returns to 0.76–0.79 (Table 2). This is expected, because the clean output alone already recovers 0.776 on Adult and 0.768 on HMDA (Experiment 2). The defenses make no claim about outputs; the measurement shows that a feature defense cannot be credited with protection when the original output is sent with the features.

6.3 Withholding the logit offset

The common logit offset changes no probability, decision or task loss, yet the frozen heads release it as part of their full logits. Withholding it lowered recovery by 0.072 (Adult) and 0.067 (HMDA) AUC. The effect appeared on 5 of the 6 primary encoder seeds and was absent on the collapsed HMDA seed 1. Across the 14 stored pairs it was established on 5. It was not established for any of the six multiclass Adult pairs, but it passed for the five-class HMDA pricing head with race. For a refitted logistic-regression head whose logits are log-probabilities, the offset is a function of the margin, so the effect is zero by construction; that head is about twice as useful (gain 0.083), and its margin alone recovers 0.773. The result is therefore a measurement

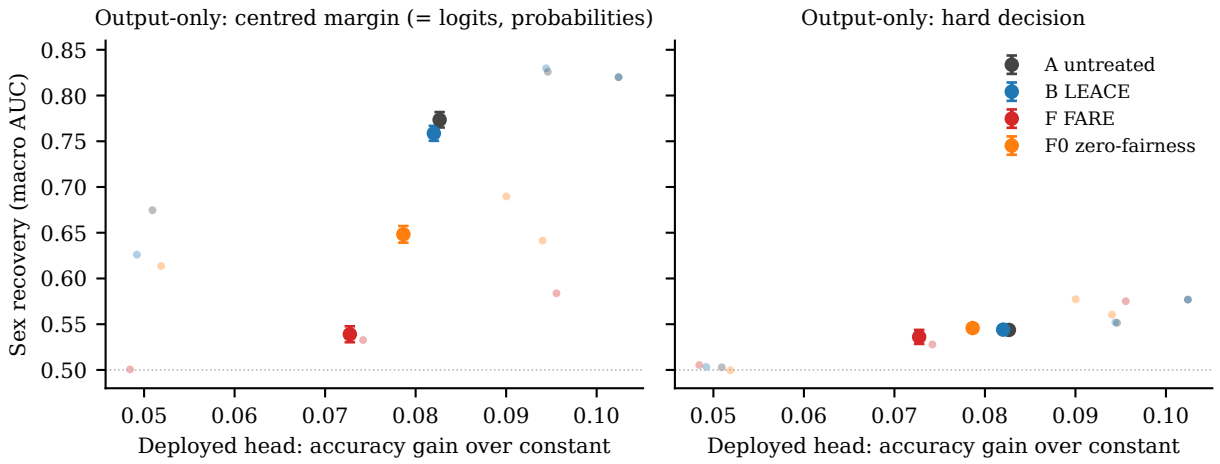


Figure 2: Useful-head comparison on Adult income/sex. Each treatment releases the output of a deployable logistic-regression head fitted on its own features. Left: the centred margin, which carries the same information as the head's logits and probabilities; right: the hard decision. Horizontal axis: the deployed head's accuracy gain over the constant; vertical axis: sex recovery (macro AUC). Large markers: means over three encoder seeds with marginal 90% intervals (the simultaneous intervals of the registered endpoints are in Table 3); small pale markers: individual seeds. The dotted line is chance.

of an output contract for particular frozen heads, and it does not amount to a general method: it is specific to these frozen heads and absent by construction for heads whose logits carry no free offset; services that return raw logits may want to check for it.

6.4 Low decision-level recovery and usefulness

The frozen heads' hard decisions leak little (0.513 on Adult, 0.505 on HMDA), but those decisions are weakly useful. The Adult frozen head gains 3–5 accuracy points over a constant and recalls 14–22% of high earners. HMDA underwriting usefulness is not established (lower bound 0.0098 against a target of 0.01). The HMDA fair-lending head predicts the same class for everyone, although its scores still vary and are informative about race and sex (centred-logit recovery 0.646 and 0.655). Low decision-level recovery here coincides with decisions that barely beat a constant. This matters for any reading of hard decisions as a defense: on these heads the low recovery cannot be separated from the low usefulness, which is why Experiment 4 repeats the comparison with useful heads.

6.5 Matched useful-head comparison

Design. On Adult income/sex, every treatment releases the output of a logistic-regression head from one common family (regularisation grid fitted on `defense_fit` minus a holdout, selected on holdout log loss) whose only runtime input is that treatment's features. The heads, LEACE maps and FARE trees are reused unchanged, and so, after checking their hashes, are four surfaces from Experiment 2: full logits alone, features alone, features with full logits, and the bypass. Experiment 2 had already reported their levels in exploratory tables. The centred, probability and hard surfaces and the banks are new and were fitted after the protocol was registered (3,924 model fits). Because the heads emit log-probabilities, their full logits, centred margin and probabilities are functions of one another and carry the same information (verified to 10^{-12}). The ten primary recovery endpoints use the centred and hard formats; the other nine primary endpoints concern utility.

Utility. All four heads are useful (Figure 2, Table 3): accuracy 0.832 (A), 0.831 (B), 0.822 (F) and 0.828 (F0) against a constant of 0.749, with balanced accuracy 0.72–0.73 and about half of high earners recalled. LEACE and the zero-fairness tree are non-inferior within one point. FARE is not established as non-inferior

Table 3: Primary useful-head endpoints (19; Bonferroni $z = 3.008$). Acc is the deployed head's accuracy and R the nonlinear attacker's output-only recovery; every recovery row has the target 0.02. Intervals are simultaneous two-sided 95% Bonferroni normal intervals; for gain and retention only their lower ends are shown. n.e.: not established; "excl. 0": the lower bound is above zero but not above the target (here the whole interval lies below 0.02).

Endpoint (target)	Point	Interval	Verdict
$\mathrm{Acc}(B) - \mathrm{Acc}(A)$ (> -0.01)	-0.0006	$[-0.0016, 0.0003]$	pass
$\mathrm{Acc}(F) - \mathrm{Acc}(A)$ (> -0.01)	-0.0099	$[-0.0152, -0.0046]$	n.e.
$\mathrm{Acc}(\mathrm{F0}) - \mathrm{Acc}(A)$ (> -0.01)	-0.0040	$[-0.0086, 0.0005]$	pass
Gain over constant, B / F / F0 (> 0.03)	0.082 / 0.073 / 0.079	$\geq$0.067 / 0.057 / 0.063	pass
Retention, B / F / F0 (> 0)	0.016 / 0.007 / 0.013	$\geq$0.013 / 0.001 / 0.007	pass
Centred: $\mathrm{R}(A) - \mathrm{R}(B)$ (> 0.02)	0.015	[0.011, 0.019]	n.e. (excl. 0)
Centred: $\mathrm{R}(A) - \mathrm{R}(F)$	0.234	[0.217, 0.252]	pass
Centred: $\mathrm{R}(A) - \mathrm{R}(\mathrm{F0})$	0.125	[0.112, 0.139]	pass
Centred: $\mathrm{R}(B) - \mathrm{R}(F)$	0.220	[0.201, 0.238]	pass
Centred: $\mathrm{R}(\mathrm{F0}) - \mathrm{R}(F)$	0.109	[0.092, 0.126]	pass
Hard: all five contrasts	-0.002 to 0.010	lower bounds < 0	n.e.

(lower bound -0.015), because seed 2 loses 2.0 points; it passes the gain and retention criteria, the latter narrowly.

Recovery from useful scores. The untreated head's score recovers sex at 0.773 and the LEACE head's at 0.759, a smaller difference of 0.015 that lies below the registered target and is carried by seed 0 alone. FARE's head recovers 0.539 and the zero-fairness tree's 0.648. Of FARE's 0.234 reduction, compression accounts for 0.125 and the fairness term for a further 0.109, and the fairness-term contrast is positive on each seed. Post hoc and descriptively, a linear attacker on the untreated head's margin recovers only 0.515, while nonlinear attackers on the same scalar recover 0.773. This is consistent with, but does not establish, the reading that the sex signal left in the score is not linearly decodable and so lies outside a linear eraser's scope.

Recovery from useful decisions. When only the decision is released, every treatment recovers 0.536–0.546 and no contrast is established. The decision has the same accuracy as the score it comes from. Post hoc and descriptively, the untreated head's decisions leak about as much as FARE's scores (difference $+0.005$, 90% interval $[-0.003, 0.013]$) while being about one point more accurate.

Complete release. With features plus own output, recovery is close to that of the features alone (A 0.816, B 0.817, F 0.550, F0 0.646), as expected, because the own head reads only those features. No offset effect was established for these heads, and none is possible in information terms, because their offset is a function of the margin. Adding the clean output to F or F0 features brings recovery back to 0.788 and 0.789.

Registered predictions. Four of the ten primary recovery predictions were correct: the contrasts of the untreated score with FARE's and with the zero-fairness tree's, and the two hard-decision contrasts that we predicted would not be established. The six wrong predictions were a larger LEACE effect on the score, a smaller fairness-term effect on the score (we predicted that $\mathrm{R}(B) - \mathrm{R}(F)$ and $\mathrm{R}(\mathrm{F0}) - \mathrm{R}(F)$ would not be established, and both passed), and three established hard-decision contrasts against the untreated head. The predictions were registered after full-logit recovery of these same heads had been reported as exploratory in Experiment 2, and the protocol did not disclose that; we disclose it here.

6.6 How much of FARE's effect is compression?

The answer differs between settings. On Adult income/sex, the zero-fairness tree's features recover 0.645 against FARE's 0.550 and LEACE's 0.812 (Experiment 2). The fairness term removes 0.109 beyond compression from the head's score (Section 6.5) and 0.096 [0.076, 0.116] from the complete release of features with the head's output (a secondary endpoint of Experiment 4, with a simultaneous interval over its 33 secondary

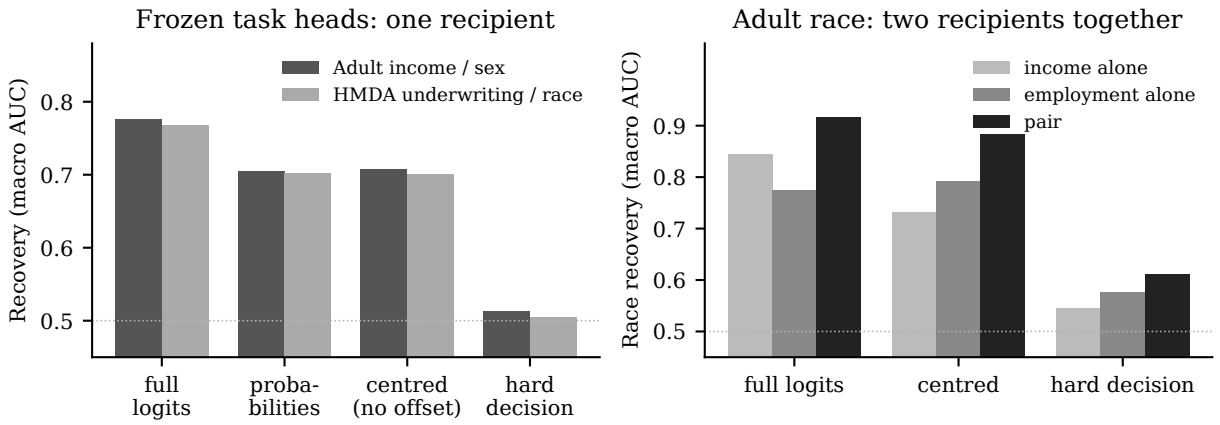


Figure 3: Left: recovery from the frozen heads' outputs by release format (Adult income/sex, HMDA underwriting/race). Right: race recovery on Adult from the income recipient, the occupation-group recipient (labelled "employment") and both linked, by format. Bars are means over three encoder seeds and show no intervals; the vertical axes start near 0.45.

endpoints). On an easier task chosen by a screen that used only validation data (Adult occupation group with age group protected; probe accuracy 0.974 for FARE and 0.975 untreated, against a constant of 0.277; the registered non-inferiority passed, -0.0004 [-0.0025, 0.0016]), FARE's complete-release recovery is 0.151 below LEACE's, but the zero-fairness tree reproduces most of that reduction: on the same complete-release surface, FARE's additional improvement is a smaller difference of 0.0074 [0.0019, 0.0129], below the registered 0.02 target. That task is easy, but it is not a recoding of a single input column, which would make the comparison trivial. An effect attributed to a fairness term therefore has to be measured against a capacity-matched tree.

6.7 Linked recipients

We evaluated one coalition: two Adult recipients, each permitted only its own task output (income; occupation group) from the untreated frozen heads, whose outputs are linked by record and pooled to recover race (macro AUC over the three Adult race groups with enough records; the other two are not estimable), which is disallowed for both (Figure 3). The linkage is granted by aligning rows, and no linkage procedure is evaluated. In this family of registered secondary endpoints, the pair beats both single recipients by more than 0.02 AUC under every format: full logits 0.917 against 0.845, centred 0.883 against 0.792, and hard decisions 0.611 against 0.576 (pair minus the better single recipient +0.034, lower bound 0.023; this is the weakest of the six registered contrasts of the pair against each single recipient under each format, all of which pass). We report absolute pair recovery and its increment over the strongest single release separately. On seed 2 under full logits, validation selected the income-only attacker for the pair, so that seed's pair-minus-income difference is exactly zero. Under the full and centred contracts each single-recipient comparator chooses among formats on validation, while the pair attacker sees only the contract's format; this asymmetry favours the single recipients in selection. These are measurements with a fixed attacker slate under perfect linkage. They show that linked outputs need an evaluation of their own; they do not show that race is carried only jointly by the two outputs. AUC is not an information measure. The pair gain does not separate evidence that each output carries on its own from evidence present only in their combination, and we prove no composition or synergy result.

7 Discussion

Where the remaining recovery comes from. LEACE's promise held wherever we could check it, FARE's certificate was too weak at our certification sizes to be checked, and the encoders' own linear constraint failed its check on 11 of 42 combinations. Apart from that failure, the recovery that remained came from attacker

classes or surfaces that the promises do not cover: nonlinear attackers on LEACE features, task outputs released beside defended features, and a second recipient whose outputs are linked to the first. This is why the three outcomes of Section 6.1 are reported separately. A method can meet its own promise while the release as a whole remains informative about the attribute.

What lowered recovery at matched utility, and what limits that reading. Among the treatments that kept the deployed head's accuracy within one point, LEACE barely changed what the score reveals and the zero-fairness tree lowered it to 0.648. FARE lowered it further, to 0.539, and its fairness-term contrast was positive on each seed. Three results limit what this supports. FARE's accuracy is not shown to be within one point of the untreated head, so its larger reduction is not established at matched utility. On an easier task the same-budget tree reproduced most of FARE's effect. And every feature defense that we evaluated together with the clean output was bypassed by it. Releasing only the decision kept accuracy and gave the lowest recovery for every treatment, at the cost of discarding scores that a recipient may need; with useful heads, decisions still recovered only 0.536–0.546, so their low recovery is not explained by weak usefulness alone.

A measured gap (post hoc, descriptive). At matched accuracy (within one point, established), the lowest score-level recovery among the evaluated treatments is the zero-fairness tree's 0.648, while the untreated decision, which has the same accuracy and no scores, recovers 0.544. The tree's own decisions recover 0.546, so for our attackers its scores add about 0.10 AUC over their thresholded decisions, in this setting. A method that released calibrated, useful scores with recovery near the decision's level would close it. We do not claim that such a method exists, nor that it is impossible.

What the findings ask of future methods. A method that aims to limit what recipients learn should state the surfaces it covers and the access model under which it is evaluated, including the attacker's labelled-data budget, and it should report recovery on the complete release beside the utility of the released output. An effect attributed to a fairness term should be measured against a capacity-matched control, and a service whose recipients share people should also be evaluated on linked releases. These are requirements for evaluating a method; they do not show that any particular method fails them.

8 Limitations, reproducibility, ethics and LLM assistance

Limitations. The evidence covers two datasets, a small number of primary settings and three encoder seeds, and the encoder seeds differ strongly (the HMDA encoder with seed 1 is collapsed). All assessment rows were used before, so nothing here is fresh confirmation. Intervals are conditional on fitted heads and attackers and omit the variation from retraining encoders. FARE is applied to stored representations rather than raw inputs, so its trees partition the encoders' representations. The two smallest HMDA race groups have too few records to enter the macro AUC. FARE's configuration and the noise scale were selected with the same attacker slate that scores them. The threat model of Section 3 is one of several, and the threat models listed there are untested. Several observations in Section 6.5 are post hoc and labelled as such.

Reproducibility. For every experiment the protocol, code version, input hashes, list of fits and endpoint families were committed before fitting. Result tables, inference code and the independent replay are in the anonymous repository; per-record outputs, weights and checkpoints are kept privately with hash indexes. The encoders' public code still exposes an earlier bound from linear R^2 to classifier accuracy that was later shown not to hold; we do not use it. The useful-head comparison used about 0.25 CPU-hours on a laptop.

Ethics. We study recovery of declared protected attributes from public benchmark data to inform release policies. Attack code is standard; we release no per-person predictions.

LLM assistance. A large language model assisted with code, analysis scripts, independent-replay scripts, literature checking and drafting. All numbers were produced by committed code. A replay that does not import the analysis code recomputed every registered endpoint; the post hoc diagnostics, the per-seed offset values and the refitted head's gain (each recomputed separately) and the intervals drawn in Figure 2 were not part of that replay. The authors are responsible for the content.

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